My Reflection

During this module I have learnt a range of new things. During unit 1 and 2 I went through a lecture cast of the principles of data visualisation, this showed me the basic information I needed to know about data visualisation to help me throughout the module and to be able to complete my assignments. Also during unit 2 all students had a collaborative discussion comparing Ryanair and easyJet websites, we compared their attributes, their colour schema, limitations and strengths etc. The purpose of this was to critically evaluate and apply the data visualisation to the visualisation process and the resulting presentations, and evaluate the capabilities of different visualisation tools and programming languages.

During unit 5 I completed the plotly task where I used Jupyter notebook and changed things on the graphs to observe the impacts. The purpose of this was to critically evaluate data visualisation grammar, and implement key analysis and visualisation techniques and effectively use the most appropriate visualisation tools.

During unit 6 I did a lecture cast on visualisation classification models, during this I learnt a lot about linear regression, the purpose of this was to help understand linear regression and applying logistic regression as classifiers and visualise the classes. During this unit I also learnt how to build a ROC curve, the purpose of this task was to teach me how to build a ROC curve to put into practice during my assignments.

During unit 7 I did the visualising dimensionality reduction and clustering models lecture cast, where I learnt about the principle component analysis, and its two principles. The purpose of this unit was to be able to use PCA and visualise and interpret my findings and also use K-means and hierarchical clustering methods to visualise clusters.

During unit 8 I completed the lecture cast on data visualisation with python, the purpose of this task was to be able to select the right data visualisation approach for machine learning related fields, select appropriate python libraries for a given data visualisation task and to appreciate the value of EDA in contact understanding the data structure.

In unit 9 I did a rough dashboard draft, this was to get feedback ready for our unit 12 assignment but unfortunately I missed the date on this so didn't get feedback, I will ensure next time I don't miss the deadline.

During unit 10 I went though all the videos of getting started on tableau, this was to help me understand how tableau works and how to create my dashboard.

All of these tasks enabled me to get a better understanding and have a wider range of knowledge needed to be able to complete my assignments.

During unit 8 I completed the programming assignment where I used the data given to develop statistical models to accurately predict the result of phone calls to sell long term bank deposits. I cleaned the data and created graphs to show the results using R. The purpose of this assignment was to critically evaluate and apply data visualisation to the whole of the visualisation process and the resulting presentations, and also to critically evaluate the capabilities of different visualisation tools and programming languages to help support the discovery and display of critical and valuable answers in different types of datasets. During this assignment I struggled a lot at the beginning to understand how to use R and how create the graphs, but once I figured out how to do it, it was relatively easy to complete the assignment, I watched videos and researched different websites to help me understand how to use R. I now feel confident I can use R to clean and visualise large amounts of data.

During unit 12 assignment, I used tableau to create a dashboard of the data from unit 8. During this assignment I struggled to understand how to do this, I couldn't create a graph that showed a trend line so I had to watch lots of videos to help me understand what I was doing wrong, I also struggled to get everything on my dashboard to look good and be in the right place. I found using tableau relatively easy otherwise and was easy to pick up once I understood what I was doing wrong. I am confident I can use tableau to create a dashboard again. During this assignment I also had to describe my design principles in a report to show I understand the structure of my dashboard and why I made the choices I did in designing it. I also had to record an executive summary to provide background information, describe what benefit the dashboard will have on users and the vision for the dashboard, this helped me understand more why the dashboard is used and its purpose and how it benefits companies to be able to visualise the data in this way. I added 4 different graphs, one with a trend line and one a histogram. I also added a drill down function and a filter. I had to write about these and describe how to use them. Using a dashboard

makes data easily accessible. This assignments purpose was to critically evaluate the capabilities of different visualisation tools and programming languages to help support the discovery and display of critical answers in different types of data sets, it was also to systematically implement key analysis and visualisation techniques and effectively use the most appropriate visualisation tools to effectively present and communicate complex information to help in the decision making process.

Overall through this module I have learned a lot of new skills and although I had a lot of trouble I managed to figure it out and learn from it.